EPFL - Computer Science

COM-407 Advanced Networks

Lecture 03 — October 1, 2025

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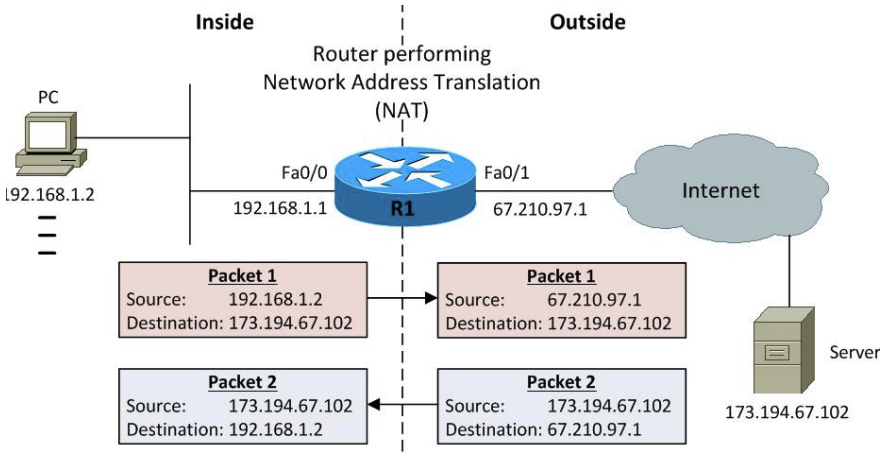
Network Layer IPv4 and IPv6 (Part 2)

How does NAT treat outbound traffic?

- When a packet goes from internal network (a.k.a LAN) to the external network (a.k.a WAN), NAT:
  - translates source IP address + port by changing the IP and UDP/TCP headers
  - stores this mapping in the NAT table
  - finally forwards the packet

How does NAT treat inbound traffic?

- When packets come from external network to internal network, NAT translates destination IP address + port
- IP forwarding is based on exact matching in the NAT table
  - If no matching entry exists, packet is dropped

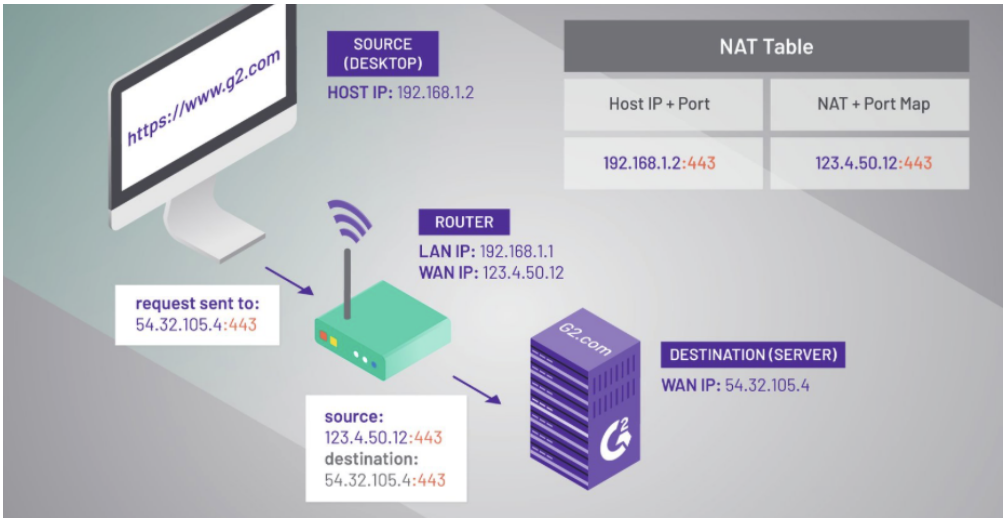


How does NAT maintain NAT table?

- It creates a table entry on-the-fly (automatically) i.e. when client on the internal network contacts server on external network
- chooses a NAT port that does not create collision in the table

Servers Behind a NAT need port forwarding

- **Problem**: Assume A runs a server at TCP port 5000; automatic operation of NAT requires communication to be started by A, which is not done for a server (it doesn't issue requests by itself)
- **Solution**: manual configuration of port forwarding in NAT
  - client connects to server at 54.32.105.4:443
  - Server needs to know its NAT IP address in advance and advertise it to potential clients
  - Server can discover its NAT IP address via STUN server or UPnP (if the server and the NAT use it)
- **Side benefit**: protection; a server port is accessed only if explicitly configured, while other ports remain unreachable.



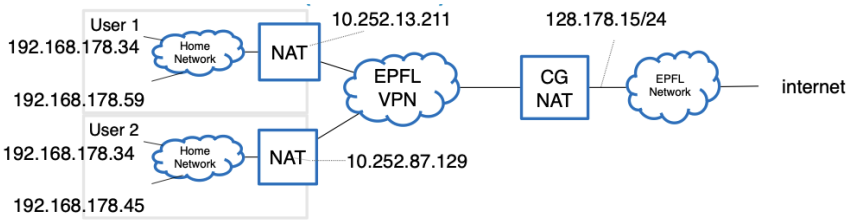
NATs and IPv6

Recall: NAT was motivated by lack of IPv4 addresses. But in IPv6, home routers often:

- do not use NAT,
  - their provider typically allocates a block of IPv6 addresses not just one as in IPv4
- provide protection by acting as a filtering router, which:
  - allows communication from outside, only if initiated from inside, unless manually configured

- Some systems still use NAT with IPv6, if local network receives only one IPv6 address, not an entire prefix
- How?
  - Use link local IPv6 addresses for internal network the internal network is a single LAN (only one subnet)
- Typical case in virtualization e.g. Virtual Box ("NAT network")

Carrier-Grade NAT (CG NAT)



What? Shares p external/NAT addresses among $\mathbf{n} > \mathbf{p}$ internal hosts

- e.g.: VPN access of EPFL uses the block 128.178.15/24 (i.e p = 256);
- VPN user 1 may appear in the public internet with address 128.178.15.x, while VPN user 2 may appear with the same IP address or not

- Some internet service providers use it, in order to reduce the number of public IPv4 addresses allocated to end-users ⇒ they use block 100.64/10 instead of 10/8
- Figure implies "NATs behind NATs":
  - e.g. User 1 appears as 192.168.178.34 at home, as 10.252.13.211 at EPFL's VPN and as 128.178.15.x in the public internet

Other NAT types

traversibility becomes harder

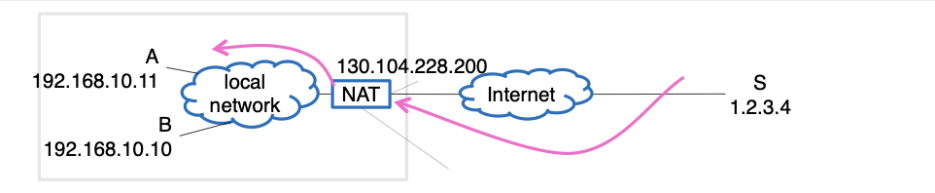
	Mapping (translation)	Consistency	Inbound access rule
Full-cone NAT	Internal IP:port ↔ NAT IP:port	Fixed across destinations	Once mapping exists, packets from *any external host* sending to the public IP:port are allowed
Address-restricted Cone			Only packets from the same **IP address** contacted by the internal host are allowed (any port)
Port-restricted Cone			Only packets from the same **IP+port** contacted by the internal host are allowed
Symmetric NAT	[Internal IP:port, Dest. IP:port] ↔ NAT IP:port	**Dynamic per-destination**: mappings vary for each external host	Only the contacted destination **IP+port** can reply back, and mapping is unique per destination

- ICMP packets don't have port numbers. What to do?
  - Some NATs just don't support ICMP; other manipulate the ICMP echo request ID
- A full cone NAT is one where all requests from the same internal IP address+port are always mapped to the same NAT IP address+port

  $$(\text{internal addr+port}) \Rightarrow (\text{NAT addr, NAT port})$$

  So, if the same host sends a packet with the same source address and port, but to a different destination, a different mapping is used. I.e. same internal IP+port may get multiple different public ports. Moreover, only the external host that receives a packet can send a packet back to the internal host.
- ICMP packets don't have a port number. Some NATs don't support ICMP. Others manipulate e.g. the ICMP echo request identifier (in ping messages) as a replacement of port number.

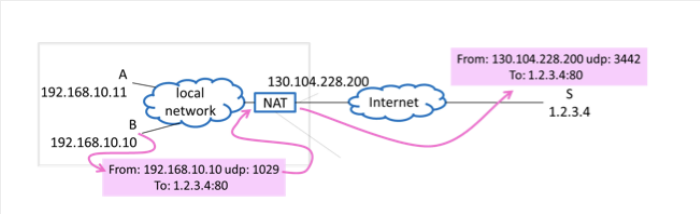
From WAN to LAN, the typical NAT may modify...



- A) The source port
- B) The destination port
- C) None of the above
- D) I do not know

Solution: B

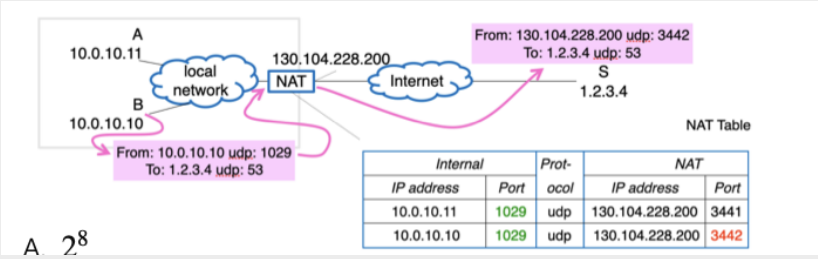
When a typical NAT has a packet to forward and an entry exists in the NAT table...



- A) NAT looks the NAT table for a longest prefix match
- B) NAT looks the NAT table for an exact matching entry
- C) None of the above
- D) I do not know

Solution: B

Suppose that the internal subnet uses a private block of addresses 10.0.0.0/8, what is the max number of internal hosts (end-systems) that a typical NAT can support?



- A) 2^8
- B) $2^8 - 1$
- C) As many as the space of UDP/TCP, 16-bit port numbers (i.e. $\leq 2 \cdot 2^{16} = 2 * 65536$)
- D) None of the above
- E) I do not know

Solution: C

- NAT creates table entries on the fly by also trying to avoid collisions on the port number. So, the max number of entries it can have is equal to the overall number of the TCP/UDP port numbers that it is allowed to use (i.e. that are not reserved for other purposes and not blocked by a firewall). This number is smaller than 2*65536, since each port number is 16bits long; hence for each of TCP and UDP, we have $2^16 = 65536 possible numbers$.
- The latter is smaller than the overall number of private IP addresses in the allocated block (which is $2^24 addresses, because the subnet mask is /8$).
- So, it is also the max number of hosts that the NAT can support—the max number of hosts is achieved by assuming that each host has only one UDP or TCP connection with some other host outside the LAN.

5. Configuration of a network interface

- A host IP interface is configured with:
  - IP address of this interface
  - Subnet mask of this interface
  - IP address of default gateway router
  - IP address of DNS server
- Can be configured manually, or automatically with:
  - IPv4 DHCP (Dynamic Host Configuration Protocol)
  - IPv6 DHCP stateful, SLAAC (stateless), DHCP stateless
- Same applies to routers connected to a provider

  - IPv4 PPP (Point to Point protocol): automatic config for telecom lines (modem, ADSL)
  - IPv6 PPP, DHCP with Prefix Delegation

DHCP (Dynamic Host Configuration Protocol)

- Client-Server app: configuration info is kept in DHCP server; host contacts the server when it needs an IP address
- Common in IPv4; also works with IPv6 (called stateful DHCP)

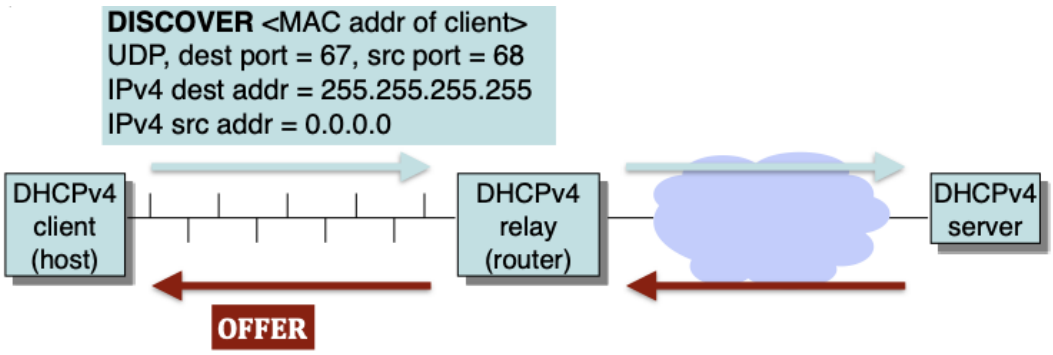
Problem: Host cannot contact the DHCP server since it does not know of any IP address initially

Solution: Use broadcast (IPv4) or multicast (IPv6) inside the LAN to discover DHCP servers

Details:

- Either DHCP server exists in the same LAN or gateway router implements a "DHCP Relay" function
- DHCP uses two phase commit with acknowledgement to avoid inconsistent reservations
- DHCP server keeps state for each "lease" (=reservation) – renews leaser after expiry

Example (IPv4):



(Self)-Autoconfiguration in IPv4

- Why? If no DHCP is available and no manual configuration is done
- What it does? Enables interconnection in unmanaged or "router-less"local networks, but not in a general setting
- How? When a host boots, it:
  - 1. picks an IPv4 link local address at random in block 169.254/16
  - 2. performs an address duplication test using broadcast IP
- Implemented in Windows, not always supported in Linux

Stateless Address Autoconfiguration (SLAACL) in IPv6

Why invented? To configure interfaces automatically without DHCP servers in IPv6

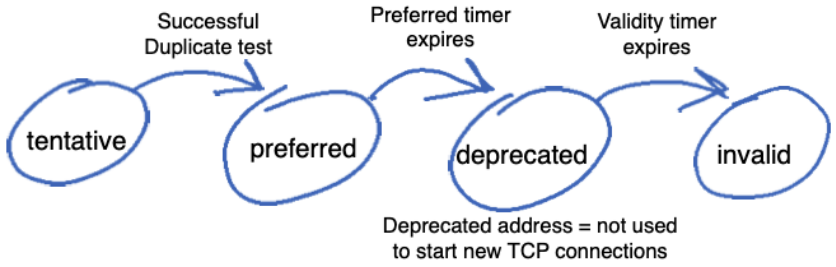
How it works?

- 1. Host auto-configures a link local address with a 64-bit prefix (fe80::/64) + a 64-bit host suffix that is derived by one of the following:
  - (i) manually, e.g. ::1
  - (ii) algorithmically from MAC address ("modified EUI 64")
  - (iii) randomly (temporary validity)
  - (iv) via "secure autoconf" (RFC7217), i.e. randomly but also related to subnet
  - (v) cryptographically with CGA
- 2. host performs address duplication test using solicited node multicast address (which has the same 24 last bits as the address it has selected) (see image for reserved address blocks for IPv6)
- 3. host computes globally valid address by obtaining the network prefix from LAN's router (via multicasting)

⇒ SLAAC is fully automatic but does not provide DNS information

Random and RFC7217 Addresses avoid Host Tracking

Solution 1: temporary, random host suffix



Solution 2: "Secure autoconf" (RFC7217), not fully random but intractable host suffix:

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Host suffix = hash (interface id, subnet prefix, secret key)|
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- Key is obtained with OS installation
- Address remains the same whenever host visits same subnet
- Address changes randomly accross different subnets

How to learn IP of DNS (after SLAAC)?

- Stateless DHCP

  - gives the IP address of the DNS server to the host
  - does not keep state for IP leases
- Router Advertisements (RFC 6106)
  - Host accesses routers in the same LAN via multicasting asking for DNS info
  - 2

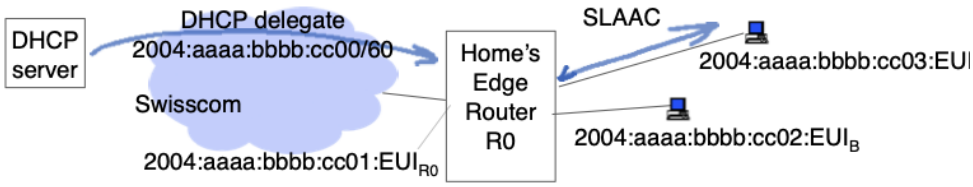
DHCP with Prefix Delegation

Why? A home (or enterprise) IPv6 router is configured by ISP using DHCP; local devices are autoconfigured using e.g. SLAAC.

And home router needs to advertize an IPv6 prefix for the entire home network

How?

- ISP DHCP server (delegating router) provides the home router with not just a single IPv6 address but also the network prefix that this router can *delegate* to its devices
- If the prefix used for the link from ISP to the home router is a subset of the delegated prefix, then it is *excluded* from the delegation (RFC 6603).
  E.g. 2004:aaaa:bbbb:cc01/64 is excluded from the delegated prefix 2004:aaaa:bbbb:cc01/60.



Question: When an IPv4 host uses DHCP, which of the following information does it acquire?

1. Its own IP address
2. A subnet mask related to the subnet it belongs to
3. The address of the default gateway
4. The address of the DNS server

- A) 1
- B) 1, 2
- C) 1, 2, 3
- D) 1, 2, 3, 4
- E) None of the above

Solution: D

With SLAAC an IPv6 host has...

- A) A link local address and, if a router is present in the subnet, also a global unicast address
- B) If a router is present in the subnet a global unicast address and no link-local address
- None of the above
- I do not know

Solution: A

Private vs link local addresses

Private addresses are allocated administratively (i.e., by a local network administrator, either statically, or automatically allocated at a single point by a suitably configured DHCP server).

Link-local addresses are allocated automatically (e.g. via SLAAC) when a computer has not been configured with a static IP-address and cannot find a DHCP server.

- Why do we need a separate space for link local?

So that the link-local addresses cannot possibly conflict with those allocated by a local, but temporarily unavailable, DHCP server.

- How are link local and private addresses handled by gateway routers?

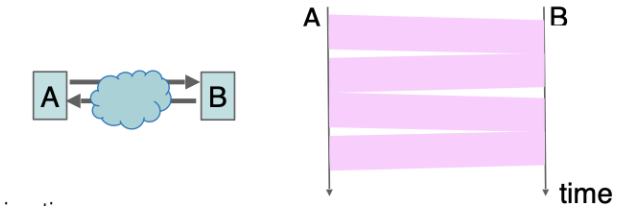
They are not routed. This is so that lots of private LANs can use the same addresses without any routing conflicts arising. Such LANs must use NAT to hide their many private addresses behind one (or a few) public addresses.

- Can both interfaces of a gateway router/NAT (internal and external) be configured with private addresses?

Yes, but they cannot belong to the same subnet. We use a different subnet at each "side" of the router/NAT.

6. Revisit IP Header: Hop Limit (HL) / Time to Live (TTL)

Why? avoid looping packets in transient loops. Transient loops may exist due to change to forwarding tables + propagation latency.



If propagation time is small compared to transmission time, a single packet (or a few packages) caught in a loop can congest the line.

How?

- Every IP packet has a field on 8 bits (from 0 to 255) (called Hop Limit in IPv6 / Time To Live in IPv4) that is decremented at every IP hop
- Every router or NAT decreases HL/TTL, switches do not
- When HL/TTL reaches 0, packet is discarded
- At source, value is 64 in principle

Traceroute leverages the HLs/TTLs

- It sends a series of packets (using UDP) to a destination, with TTL = 1, 2, 3, ...
- Routers on the path discard packets and send ICMP error message back to source
- Source learns address of router on the path by looking at source address of error message
- tracert (windows) is similar, but uses ICMP

Checksum

- IPv4 only, protects header against bit errors
- absent in IPv6 (layer 2 and router hardware assumed to have efficient error detection)

A host generates a packet with Hop Limit = 1

- A) This packet is invalid
- B) This packet will never be forwarded by a switch nor by a router
- C) This packet will never be forwarded by a switch but may be forwarded by a router
- D) This packet will never be forwarded by a router but may be forwarded by a switch
- E) None of the above is true
- F) I do not know

Solution: D

This packet cannot be forwarded by a router because it would decrement the HL and obtain 0. It can be forwarded by a switch because a switch does not examine the IP header.
Note that such a packet is perfectly valid. Sources put HL=1 when they want to be sure that the packet remains in the LAN.